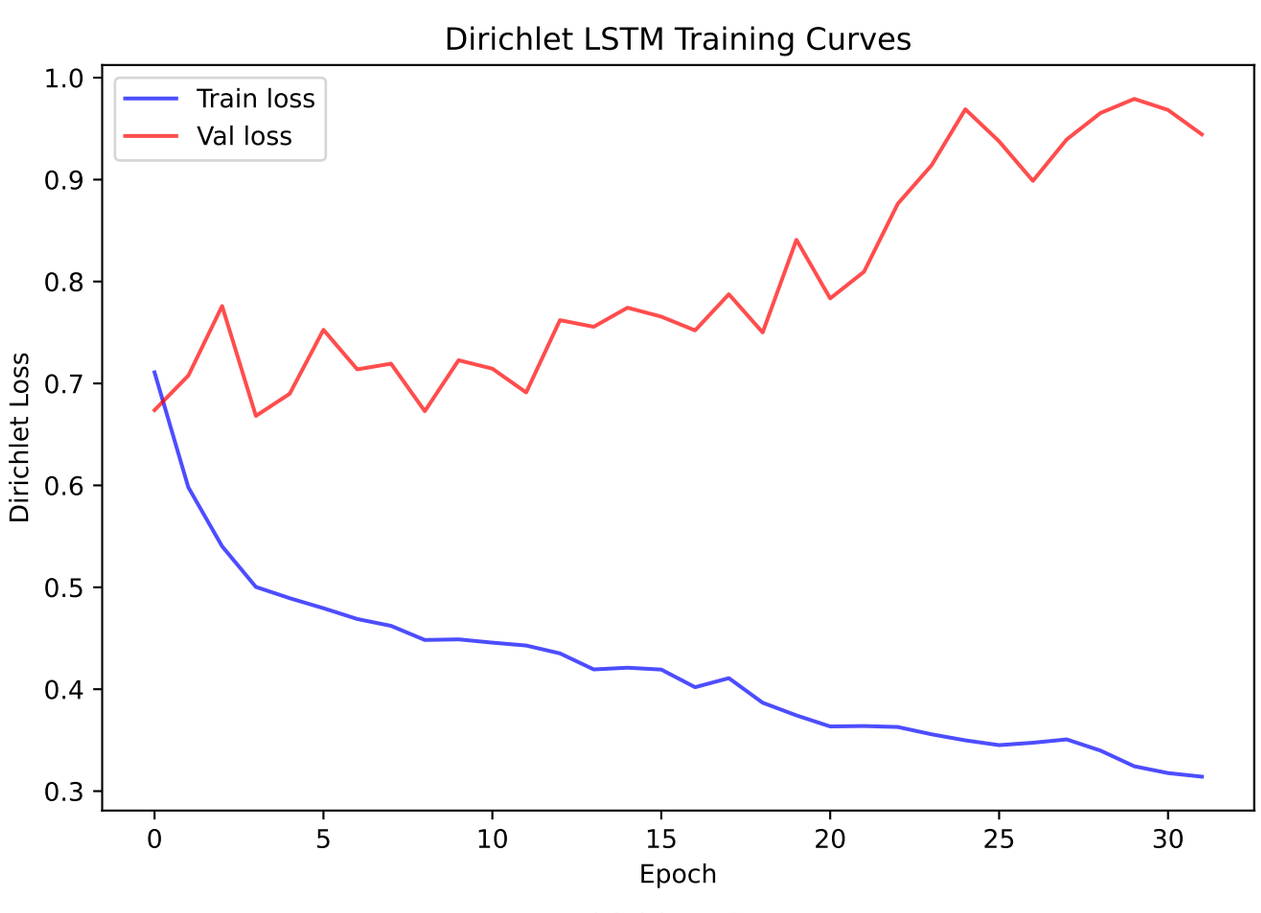
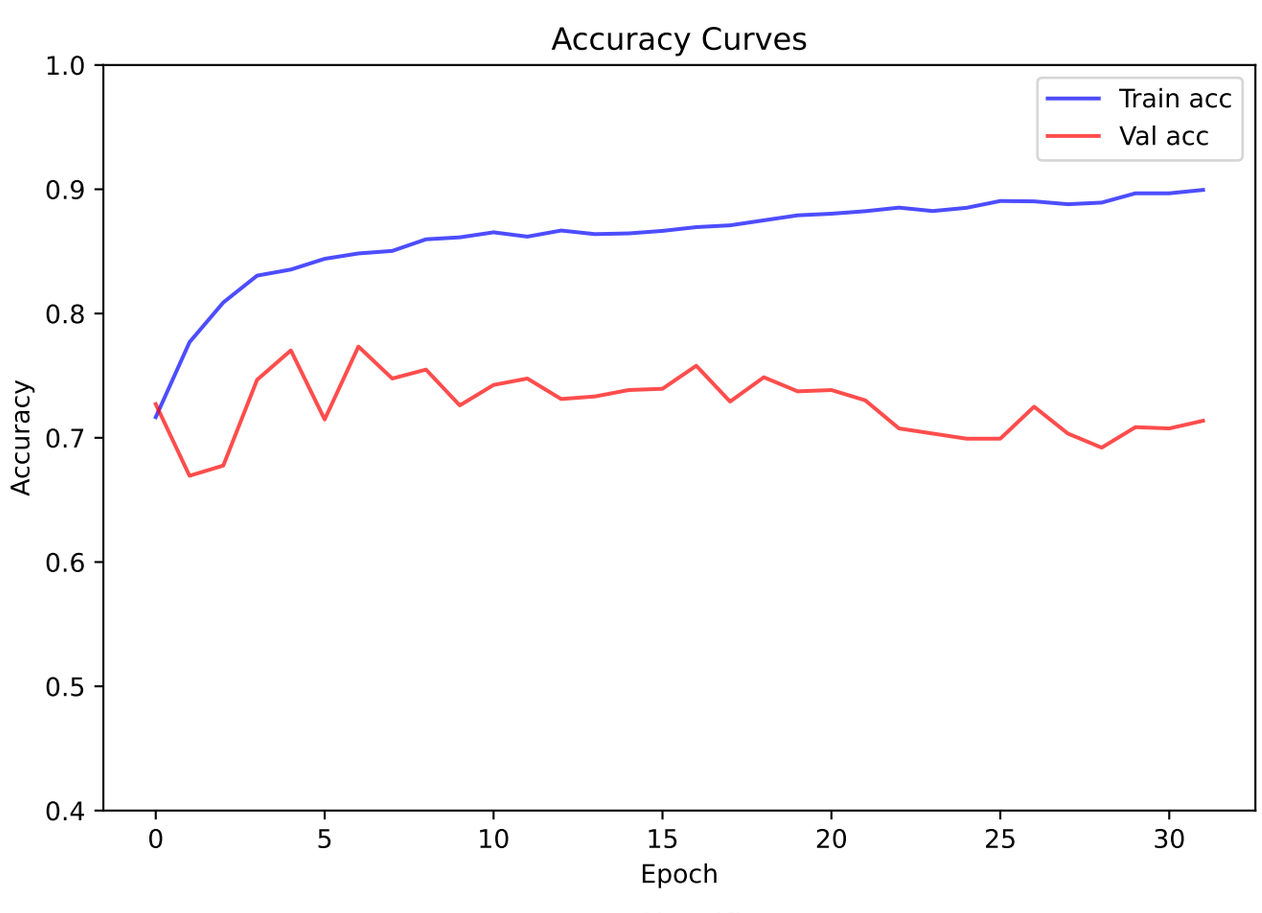
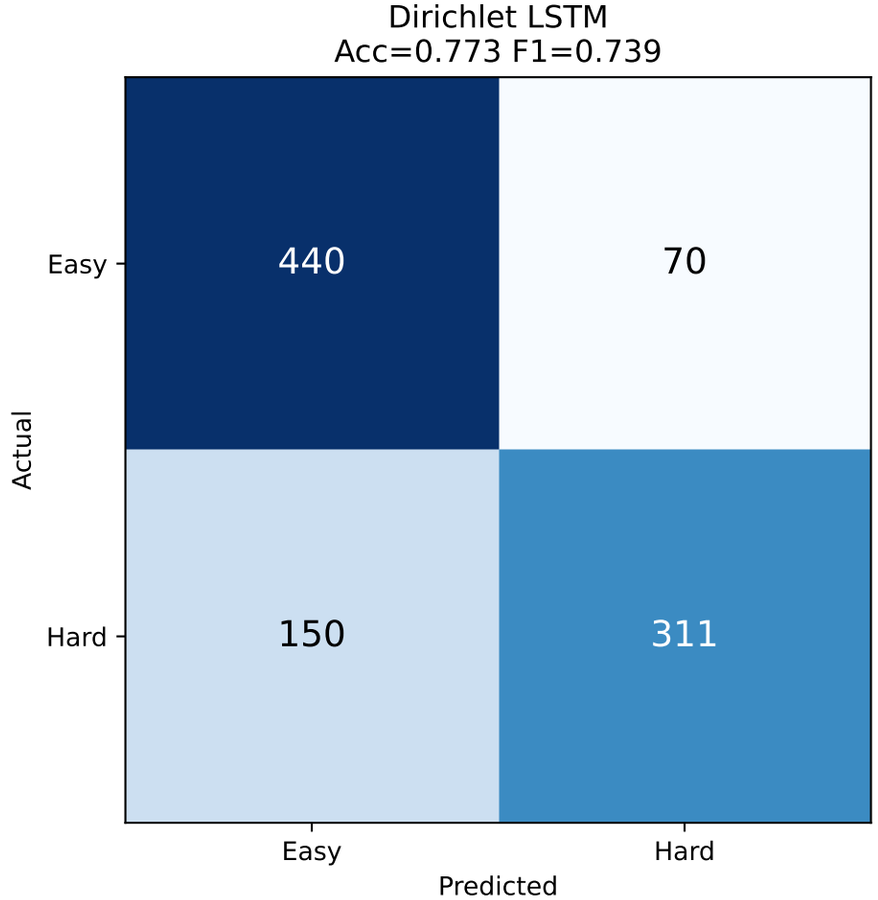
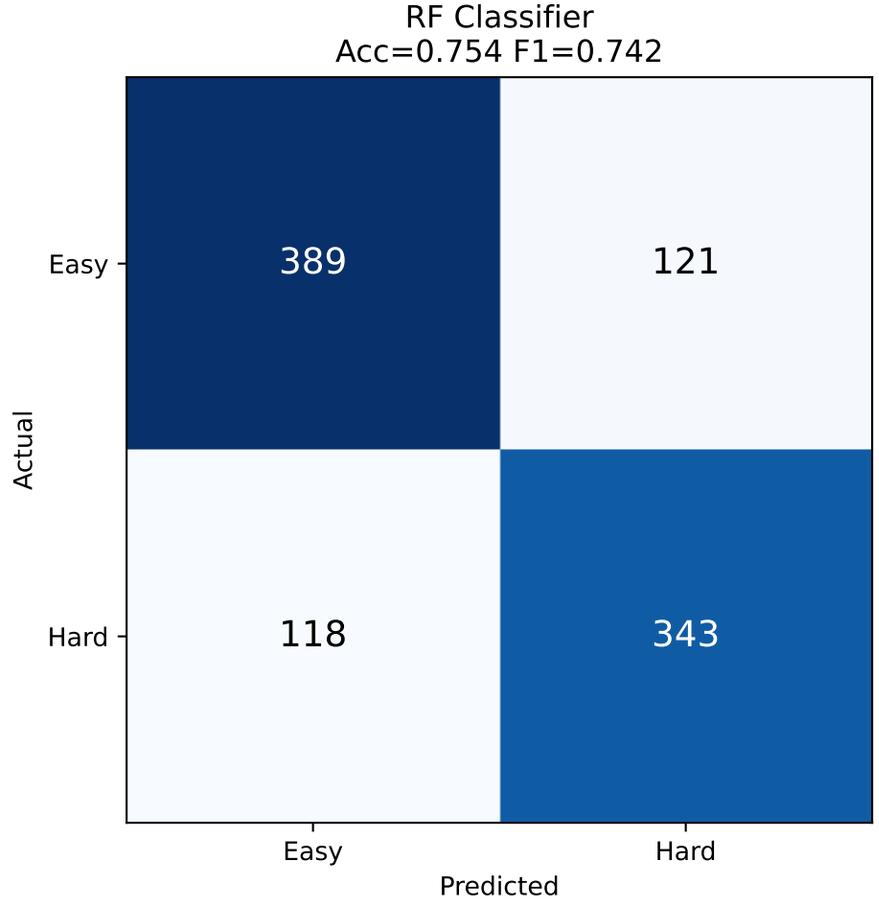
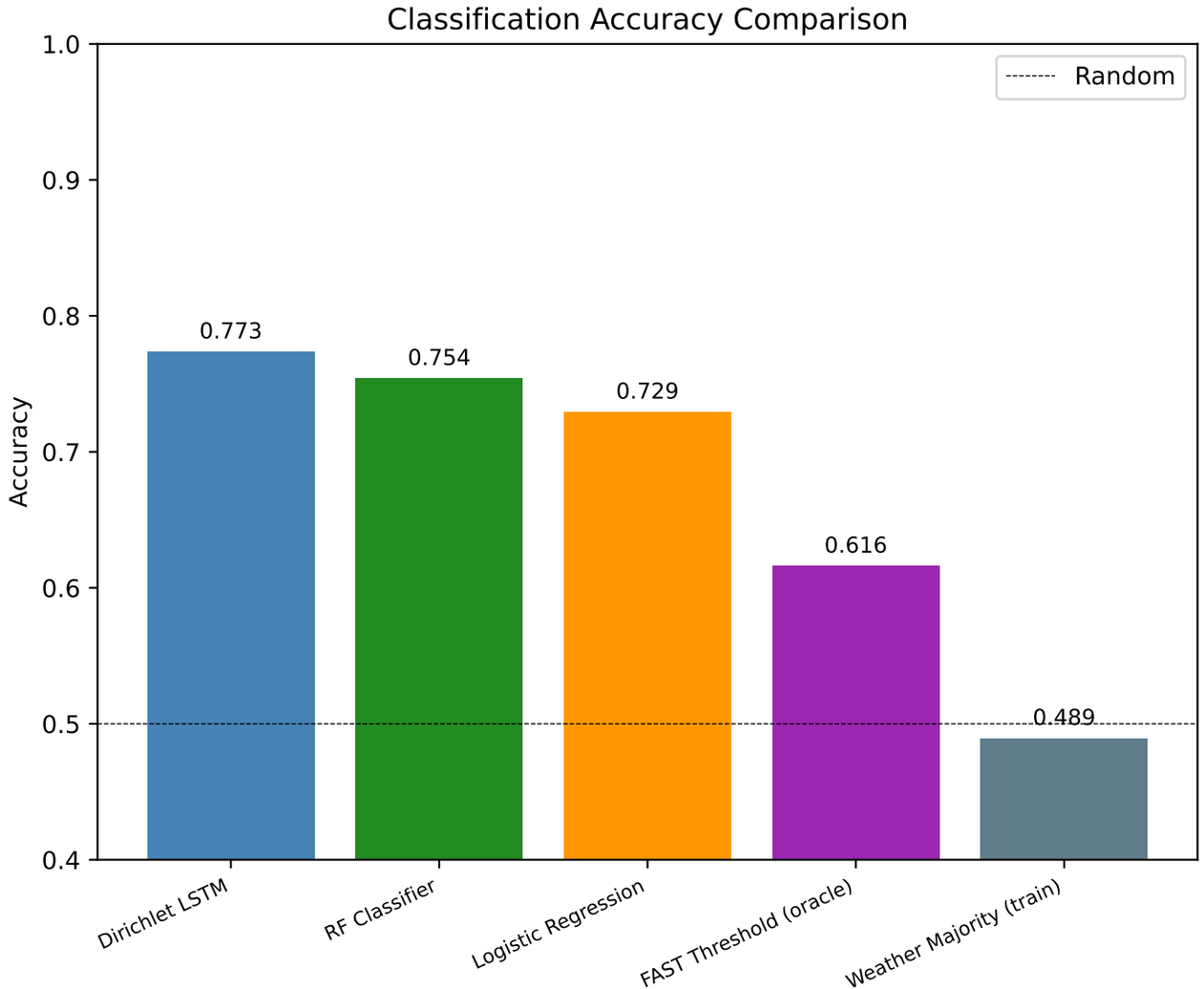
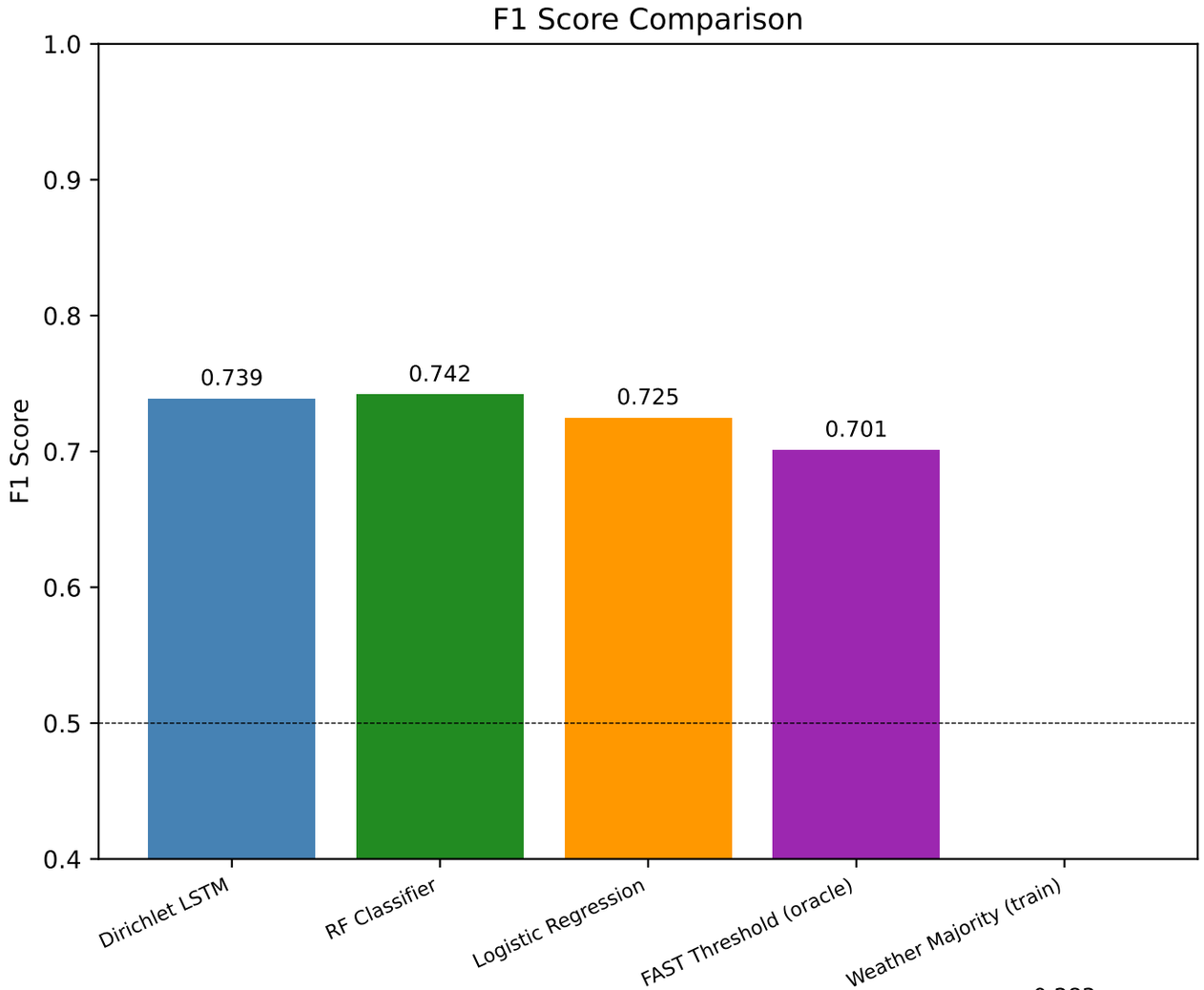

Dirichlet LSTM Training Curves
Train loss
Val loss
Dirichlet Loss
Epoch
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
5
10
15
20
25
30

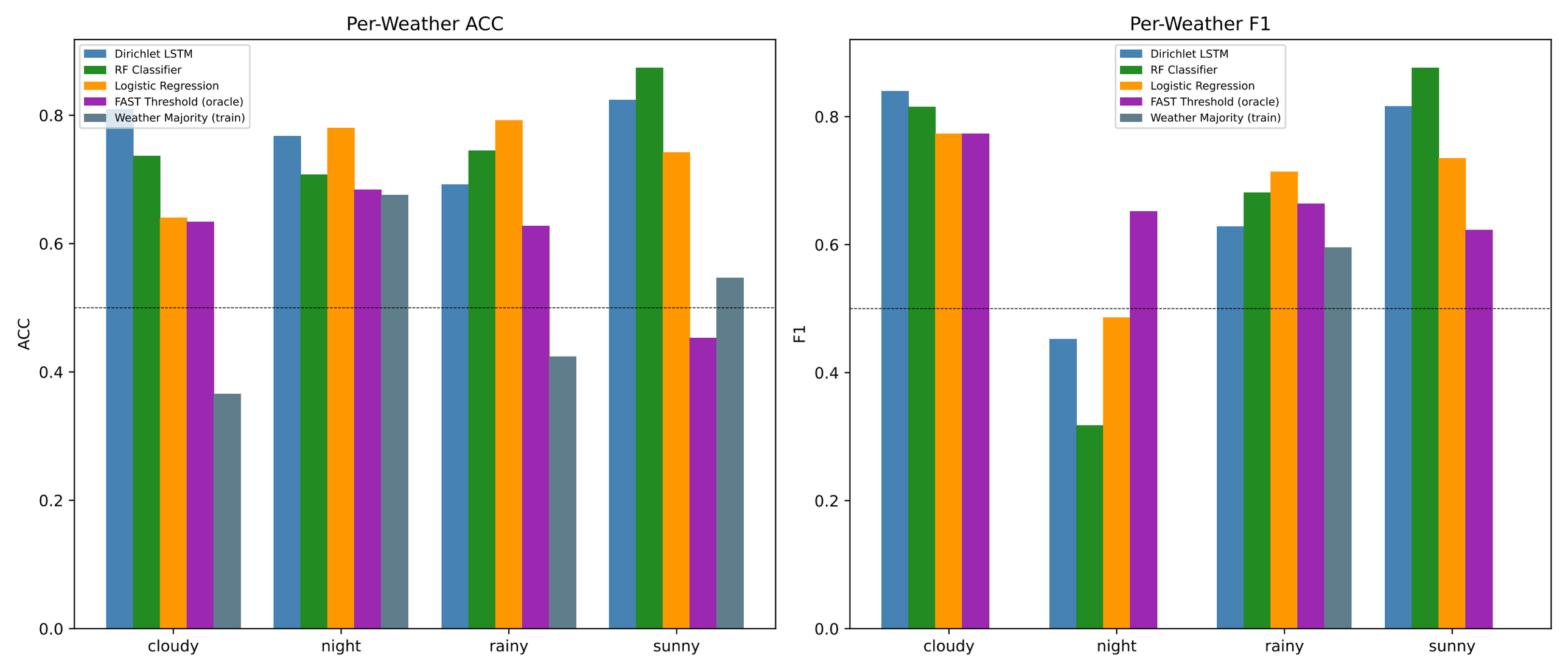

Accuracy Curves
Train acc
Val acc
Accuracy
Epoch
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
5
10
15
20
25
30

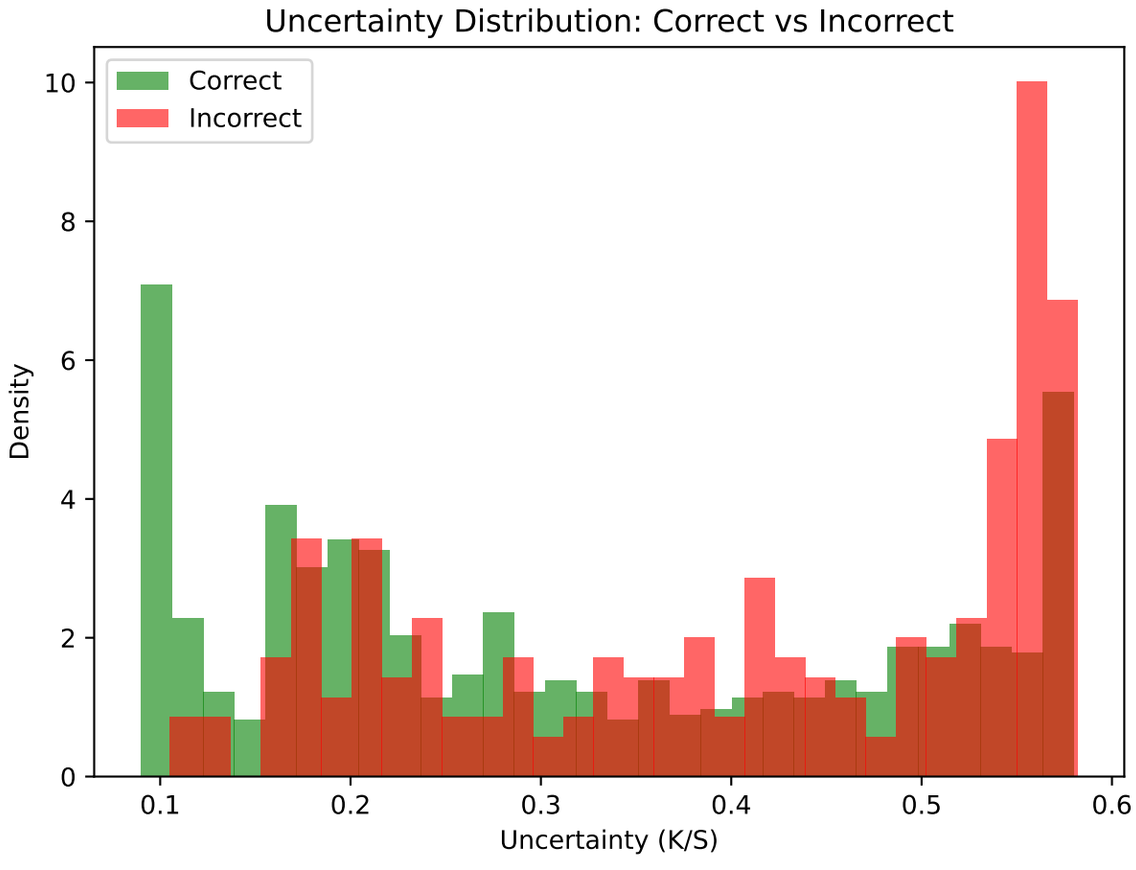

Dirichlet LSTM
Acc=0.773 F1=0.739
440
70
150
311
Easy
Hard
Actual
Predicted

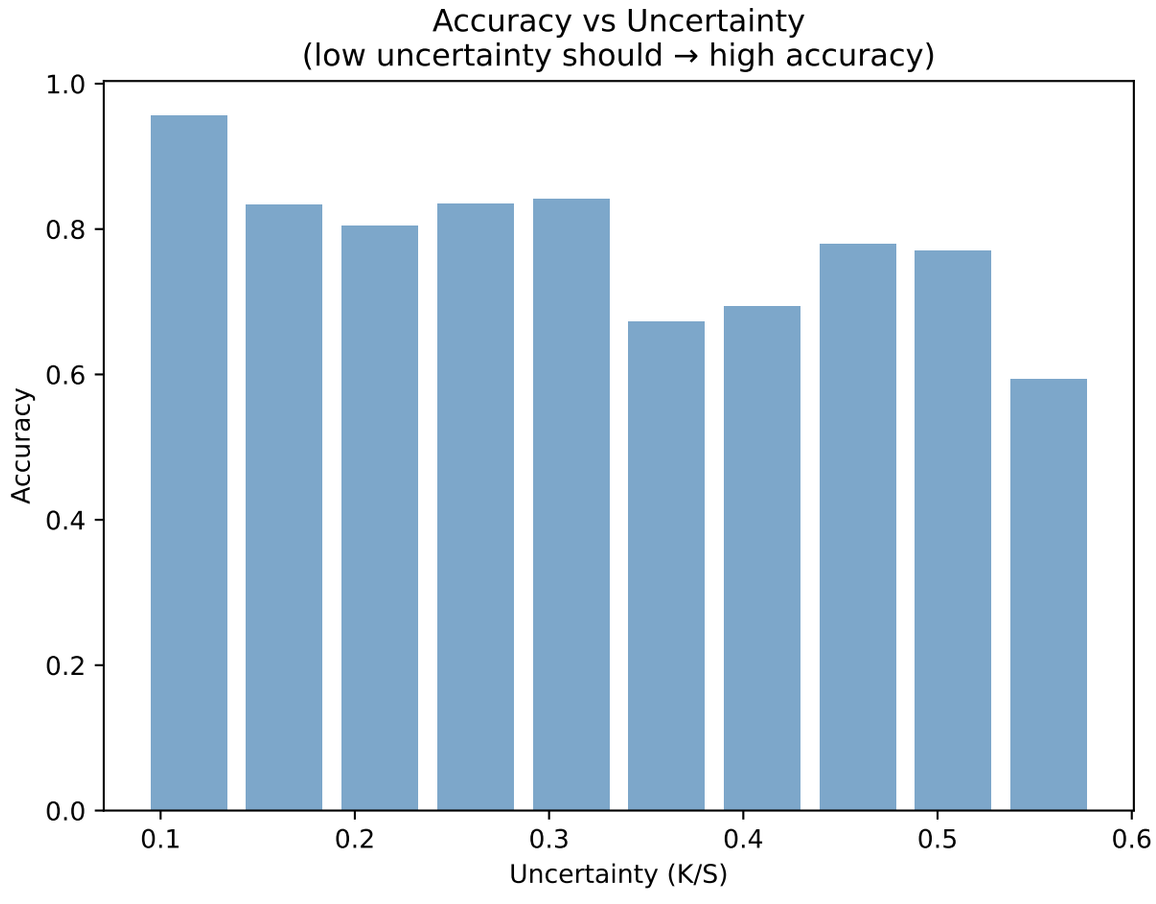

RF Classifier
Acc=0.754 F1=0.742
389
121
118
343
Easy
Hard
Actual
Predicted

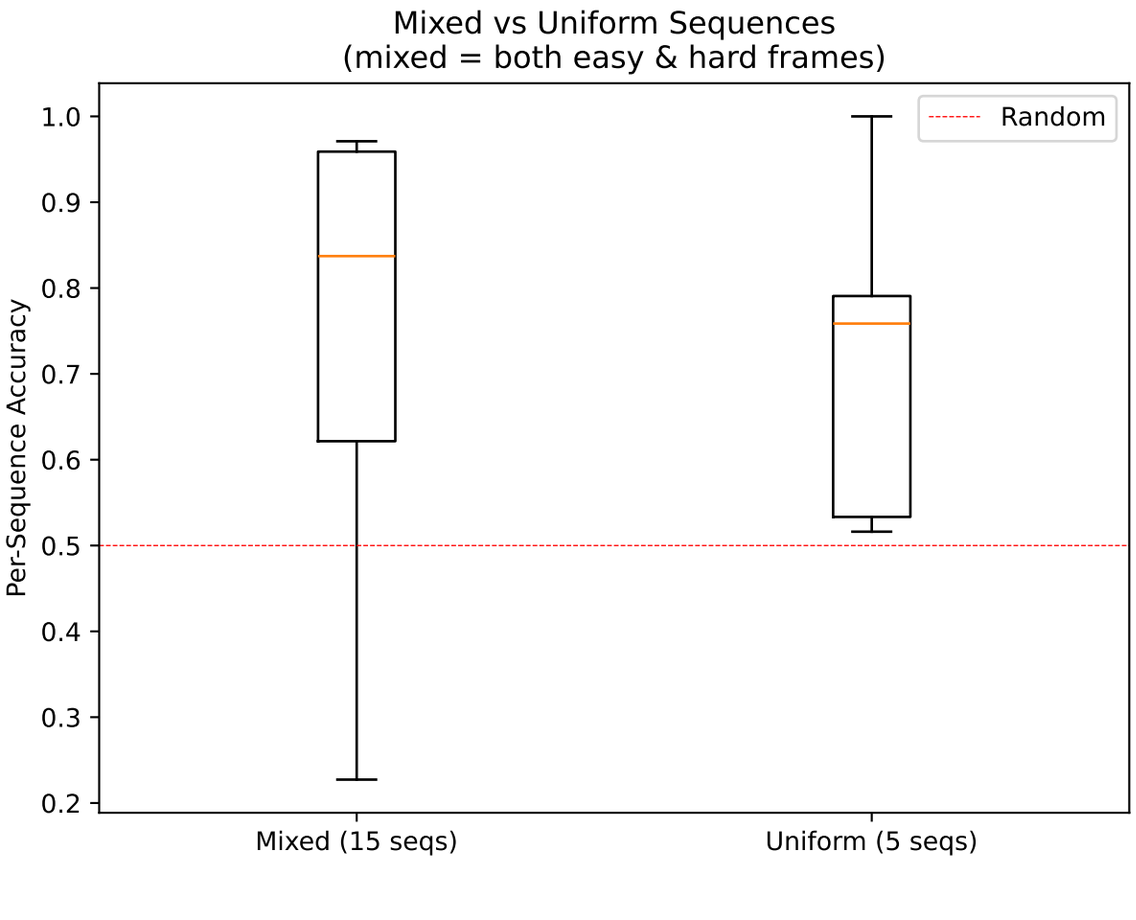

Classification Accuracy Comparison
Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Random
0.773
0.754
0.729
0.616
0.489
Dirichlet LSTM
RF Classifier
Logistic Regression
FAST Threshold (oracle)
Weather Majority (train)

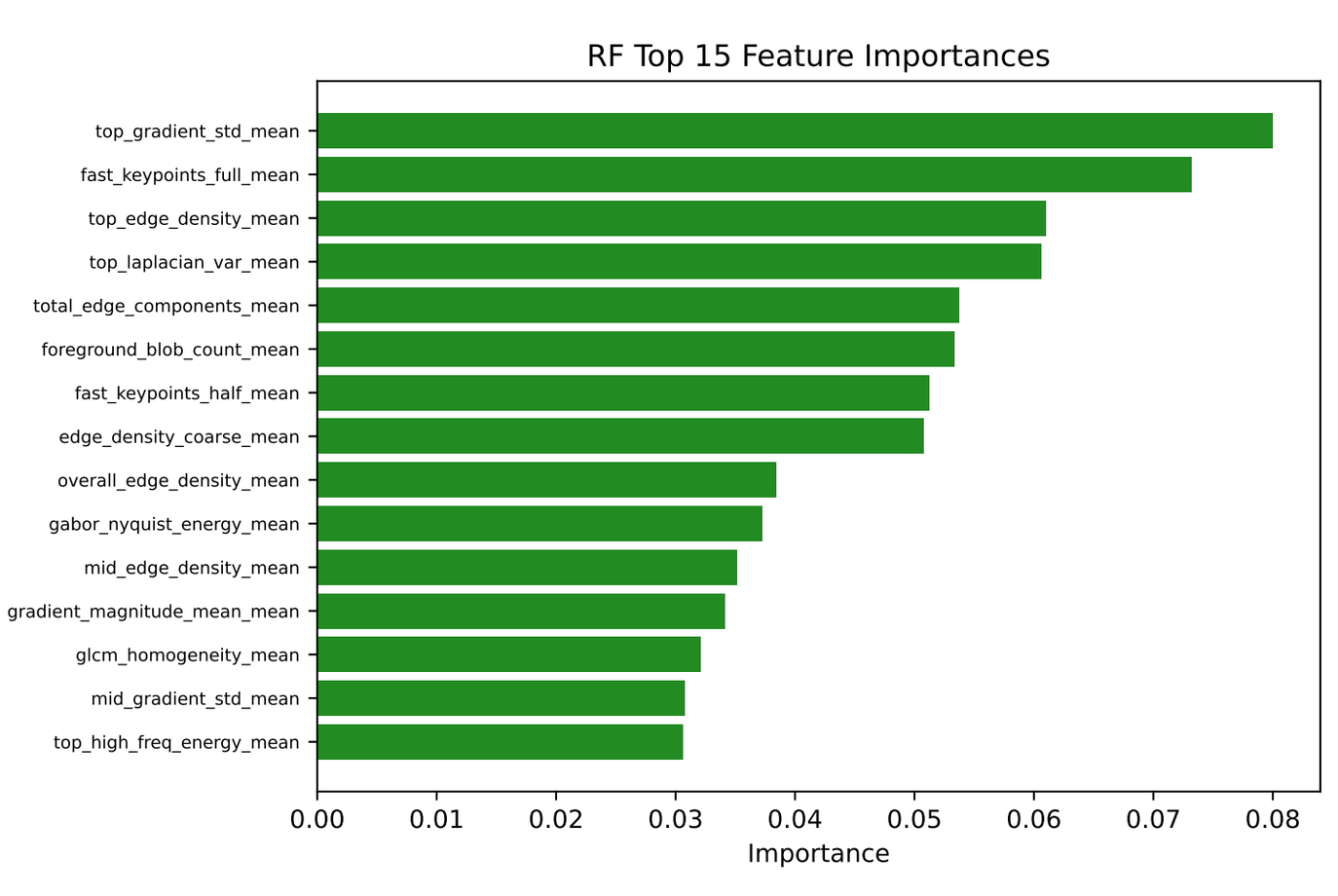

F1 Score Comparison
F1 Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.739
0.742
0.725
0.701
0.283
Dirichlet LSTM
RF Classifier
Logistic Regression
FAST Threshold (oracle)
Weather Majority (train)


Per-Weather ACC
Dirichlet LSTM
RF Classifier
Logistic Regression
FAST Threshold (oracle)
Weather Majority (train)
ACC
0.0
0.2
0.4
0.6
0.8
cloudy
night
rainy
sunny
Per-Weather F1
F1


Uncertainty Distribution: Correct vs Incorrect
Correct
Incorrect
Density
Uncertainty (K/S)
0
2
4
6
8
10
0.1
0.2
0.3
0.4
0.5
0.6


Accuracy vs Uncertainty
(low uncertainty should → high accuracy)
Accuracy
Uncertainty (K/S)
0.0
0.2
0.4
0.6
0.8
1.0
0.1
0.2
0.3
0.4
0.5
0.6


Mixed vs Uniform Sequences
(mixed = both easy & hard frames)
Random
Per-Sequence Accuracy
Mixed (15 seqs)
Uniform (5 seqs)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0


RF Top 15 Feature Importances
top_gradient_std_mean
fast_keypoints_full_mean
top_edge_density_mean
top_laplacian_var_mean
total_edge_components_mean
foreground_blob_count_mean
fast_keypoints_half_mean
edge_density_coarse_mean
overall_edge_density_mean
gabor_nyquist_energy_mean
mid_edge_density_mean
gradient_magnitude_mean_mean
glcm_homogeneity_mean
mid_gradient_std_mean
top_high_freq_energy_mean
Importance
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08

Dirichlet LSTM vs Baselines — Binary Classification

Model	Acc	F1	Prec	Rec	AUC	Deployable?
Dirichlet LSTM	0.773	0.739	0.816	0.675	0.820	Yes
RF Classifier	0.754	0.742	0.739	0.744	0.824	Yes
Logistic Regression	0.729	0.725	0.700	0.751	0.809	Yes
FAST Threshold (oracle)	0.616	0.701	0.556	0.948	0.000	Yes
Weather Majority (train)	0.489	0.283	0.424	0.213	0.000	Yes

```
Task: classify mean(fn_nano) over next 30 frames as easy (<2.70) or hard (>=2.70)
                        Threshold: median fn_nano = 2.70
                      Input: 30 frames x 20 features = 1 sec
                        Train stride: 15, Eval stride: 30
           Split: 80 train / 20 val sequences (stratified by weather)
         No num_gt filter, no per-seq normalization, no look-ahead bias

                    Per-sequence accuracy (Dirichlet LSTM):
                     All sequences: median=0.813, mean=0.753
        Mixed only:    median=0.837, mean=0.764, n_above_50%=12/15
```